SDC Hackathon 2026 · Domain 1: The Aiversecity Future-Ready Workforce · SDG 4.4 and 8.5

SkillFolio

Your skills are a portfolio. AI is the market.

A web app that shows Malaysian graduates which future roles fit their skills, why, and what to learn next.

1 Profile | 2 Skill Market | 3 Matches | 4 Learning plan

Team MIX MECHS · University of Southampton Malaysia

The problem

Graduates guess what to learn

66.0%

of Malaysian graduates aged 24 and under worked in jobs below their qualifications in 2024

32.2% of all employed graduates (1.6 million)

600,000+ workers must reskill within 3 to 5 years

MADE-UP USER

Meet Aina

Business-analytics graduate, Kuala Lumpur. About 200 hours to learn this year, RM 300 to spend.

Which of my skills are in demand?
Which roles can I reach?
Is this course worth my hours?

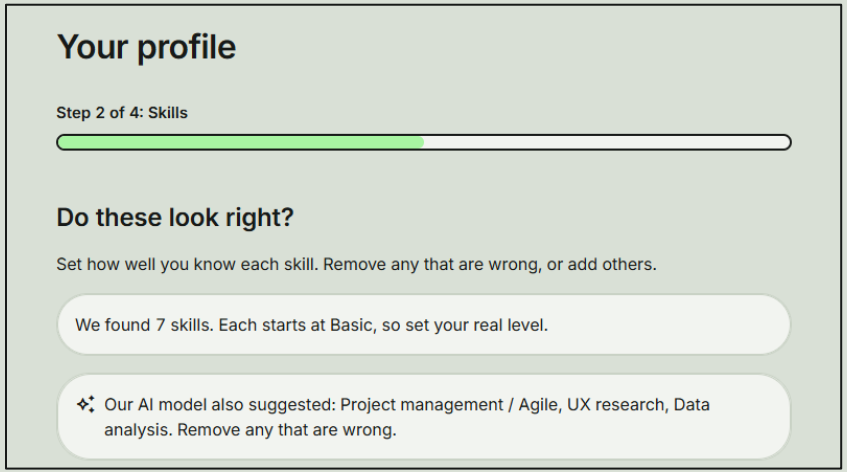
Chatbots answer, but can make up jobs, courses and numbers.

Sources: DOSM, Graduates Statistics 2024 · MOSTI minister, Malay Mail, 7 Jan 2025. Aina is a synthetic app profile.

From resume to skill portfolio

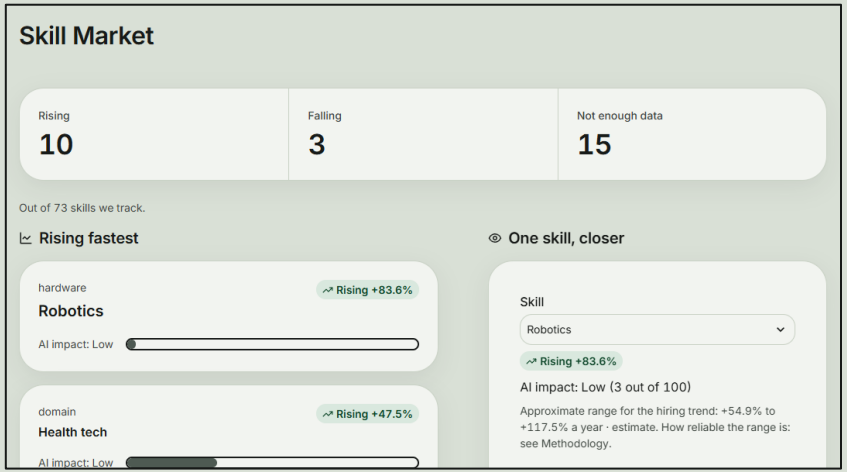
1 Profile

Paste a resume. Keywords and our AI model pick skills from a list of 73. Remove any that are wrong.



2 Skill Market

Hiring trend per skill from 7,961 job posts over 25 months, plus its AI impact.

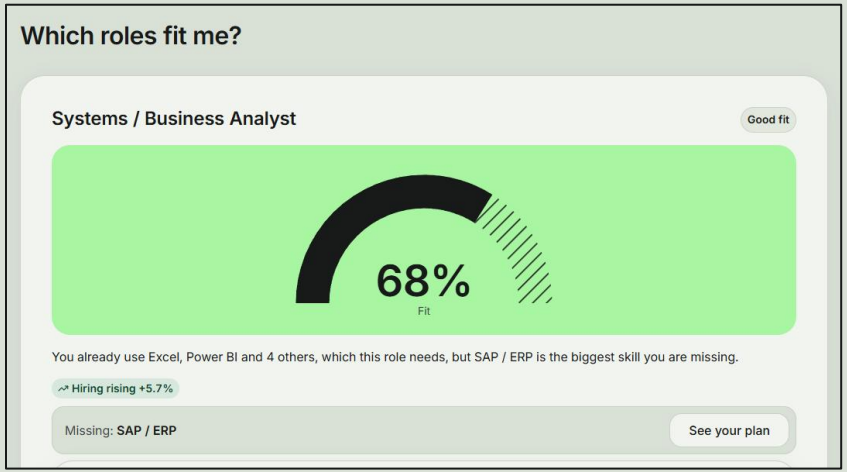


App screenshots. Job posts: Hacker News "Who is hiring", Oct 2024 to Oct 2026. AI impact: Anthropic Economic Index.

Roles that fit, and a plan

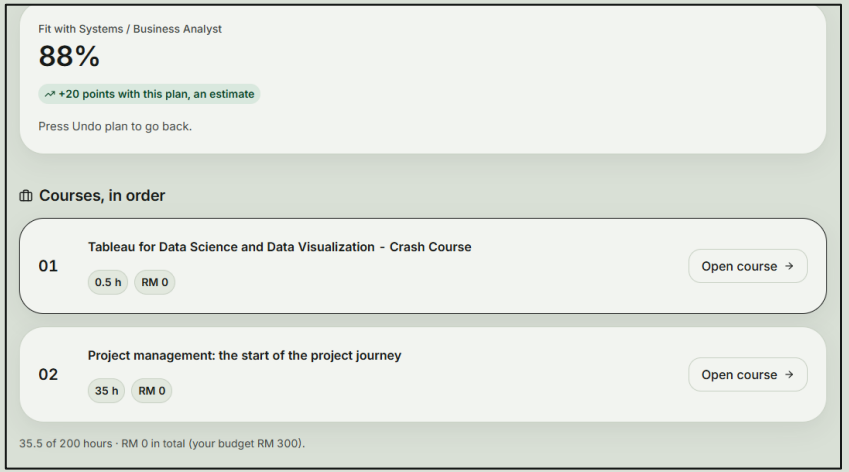
3 Matches

20 roles ranked by fit. Aina's best: Systems / Business Analyst at 68%. Biggest gap: SAP / ERP.



4 Learning plan

Two free courses, 35.5 hours, RM 0. Apply plan: fit goes from 68% to 88%, an estimate.



From the app (snapshot 23 Oct 2026, moderate AI). Aina's skills: hiring trend +9.7% a year, AI impact 19/100.

Why you can trust it

Every number can be checked

Forecast range, tested by simulation

48% → **~76%**

before — after widening

Our nominal 80% range held the true trend only 48% of the time. We widened it, and it now holds about 76%. We say so in the app.

915

automated tests on the data and the engine

73 · 20 · 52

skills, roles and free courses: the only things it can recommend

Sourced

every number shows its source and date

Estimates

forecasts are labelled as estimates, under two AI scenarios

Simulation and widening factor: docs/SPEC.md and the app's Methodology page. Test count: pytest on main, 24 Oct 2026.

The AI, and where it runs



Statistics, not a chatbot

Trend forecasts with a range, role matching and a course planner. No generative language model in the product.

Live RunPod call, 24 Oct 2026

A resume naming no tools returned SQL, Power BI, Excel in 2.5 s (warm). If RunPod is slow or down, keyword matching takes over.

Stack: Streamlit, Plotly, pandas, NumPy, Docker. Built with the help of Claude Code and Replit Agent; the team reviewed the code and ran the tests before merging.

SDG impact framework

Impact: skills for decent work

SDG 4.4

More youth and adults with relevant skills for employment and decent jobs

SDG 8.5

Full, productive employment and decent work, including for young people

Measured today

+41 points

average Fit gain from Apply plan, within each learner's hours

Phase 2 targets

≥ 40 pts

average Fit gain, with AI impact and outlook shown beside it

100%

of recommended roles, courses and numbers sourced and dated

< 3 min

from profile to plan

≥ 80%

of 10 outside testers name a reachable role and their next course

Measured 22 Oct 2026, 5 sample profiles × top 3 roles: AI impact about flat (−0.3 pts); outlook falls in 11 of 15.

Honest limits

What it can't do yet

- Tech and data only: 20 roles, 73 skills
- Job posts are mostly US and remote tech, not Malaysian
- Progress is self-reported; a short course can lift fit a lot
- A portfolio lasts one browser session
- The AI model can over-suggest, and a cold start takes up to a minute
- Forecasts are estimates under two scenarios, not predictions

Next, in Phase 2

- A Malaysian job source, so non-tech skills get real trends
- Saved portfolios
- The RunPod model reads job posts too
- Bahasa Malaysia interface

From the README, Known Limitations and Future Improvements.

Team MIX MECHS · University of Southampton Malaysia

Who built it

Jaiydaan
Team Lead, Engine + RunPod

Atheek
Data & Trends

Kuhan
App & Replit

Oscar
Research, SDG & Pitch

Your skills are a portfolio. AI is the market.

SkillFolio shows the evidence. The user makes the decision.

github.com/SDC-HACKATHON-MIX-MECHS/skillfolio